T3 D_+0: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

D_+0

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y1Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(lambdaT3ExtraM000200I1, \{\}, 0)CG(T3ExtraCGM000200I1, \{Index(d\$1, T3BSMd3), Inde}$

$\text{+Coupling(lambdaT3ExtraQ000400I1, \{\}, 0)Bar(Field(NewScalar1, Scalar, \{Index(d881, T3BSMd3)\}, \{\}))Bar(Field(NewScalar1, Scalar, \{Index(d882, T3BSMd3)\}, \{\}))Bar(Field(NewScalar1, Scalar, \{Index(d883, T3BSMd3)\}, \{\}))Bar(Field(NewScalar1, Scalar, \{Index(d884, T3BSMd3)\}, \{\}))CG(T3ExtraCGQ0004001, \{Index(d881, T3BSMd3), Index(d882, T3BSMd3), Index(d883, T3BSMd3), Index(d884, T3BSMd3)\})}$

$$+\text{Coupling}(\text{lambdaT3ExtraQ1102001I}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQ1102001}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ021010I1}, \{\}, 0))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQ021010I1}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})))$$

$$+\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{T3MixCG}, \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaH1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraM000200I1}, \{\}, 0))\text{CG}(\text{T3ExtraCGM000200I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ1102001I}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQ1102001I}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ001300I1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3ExtraCGQ001300I1}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar1SelfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Scalar1SelfCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ001300I1}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3ExtraCGQ001300I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ000400I1}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd3})\}, \{\})\text{CG}(\text{T3ExtraCGQ000400I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3}), \text{Index}(\text{d\$\$3}, \text{T3BSMd3}), \text{Index}(\text{d\$\$4}, \text{T3BSMd3})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3}, \{\}, 0))\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{Bar}(\text{T3MixCG}), \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ000211I1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{CG}(\text{T3ExtraCGQ000211I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})$$

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})))$$

$$+\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)}))\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ021010I1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQ021010I1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})$$

$$+\text{Coupling}(\text{lambda12Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ000211I1}, \{\}, 0))\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{CG}(\text{T3ExtraCGQ000211I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$2}, \text{T3BSMd3})\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd3})\}, \{\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})^2\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))^2$$

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$-\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\}))\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{T3Y2CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2}))\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})$$

$$+\text{Coupling}(\text{y1Conj}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{T3Y1ConjCG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(1)), \text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})))$$

$$+\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{CG}(\text{T3Y1CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(1)), \text{Transp}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})))$$

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\}, \{\})\text{Bar}(\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0))\text{CG}(\text{Bar}(\text{T3Y2CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(-1)), \text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})$$

$$+\text{Bar}(\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\}))\text{CG}(\text{Bar}(\text{T3Y1CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Transp}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})\text{Bar}(\text{Coupling}(\text{y1Conj}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0))\text{CG}(\text{Bar}(\text{T3Y1ConjCG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3})\})\text{NCM}(\text{Transp}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa1ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`
- `Scalar1SelfInv2`
- `Scalar2SelfInv1`
- `HiggsPortal1Inv1`
- `HiggsPortal1Inv2`
- `HiggsPortal2Inv1`
- `ScalarCrossPortalInv1`

- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_2_1_1_1
- T3ExtraQ_0_0_0_4_0_0_1
- T3ExtraQ_0_0_1_3_0_0_1
- T3ExtraQ_0_2_1_0_1_0_1
- T3ExtraQ_1_1_0_2_0_0_1
- T3ExtraM_0_0_0_2_0_0_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3\left(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ Y_e\right)_{ij} + 2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= \mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big)\,(y_1)_{ij} \\ &+ 2\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big)\,(\mathrm{y1Conj})_{ij} \\ &+ T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{12}(29\left(y_1\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\right)_{ij} + 29\left(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_1\right)_{ij} + 6\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 18\Big(\\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= 2T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 3\left(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_2\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 4\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{4}(3(\Big((y_1)^T\ (y_1)^*\ M_F\Big)_{ij} + \Big((\mathrm{y1Conj})^T\ (\mathrm{y1Conj})^*\ M_F\Big)_{ij}) + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$16\pi^2 \frac{dm_1^2}{d\ln\mu} = m_1^2(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$$

$$
\begin{aligned}
&+ \text{muT3GeneralS1Sq}(4\,\mathrm{Tr}\Big(y_1\ (\text{y1Conj})^\dagger\Big) + 20\,(\text{lambdaT3GeneralQ001300I1})^*)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)\\
&- 4\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (\text{y1Conj})^T\ (\text{y1Conj})^*\Big)\\
&+ 4\,(\text{muT3GeneralS1Sq})^*\,\mathrm{Tr}\Big(\text{y1Conj}\ (y_1)^\dagger\Big)\\
&+ 20\text{lambdaT3GeneralQ001300I1}\,(\text{muT3GeneralS1Sq})^*\\
&- 12{g_2}^2 m_1^2\\
&+ 2\lambda_{12}^{(1)} m_2^2
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = {}& m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})\\
&- 8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big)\\
&+ 12\text{muT3GeneralS1Sq}\,(\text{lambdaT3GeneralQ000211I1})^*\\
&+ 12\text{lambdaT3GeneralQ000211I1}\,(\text{muT3GeneralS1Sq})^*\\
&- 6{g_1}^2 m_2^2\\
&+ 6\lambda_{12}^{(1)} m_1^2
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = {}& \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 4\text{lambdaT3GeneralQ021010I1}\,(\text{lambdaT3GeneralQ021010I1})^*\\
&+ 24\text{lambdaT3GeneralQ110200I1}\,(\text{lambdaT3GeneralQ110200I1})^*\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 6{\lambda_{HS_1}^{(1)}}^2\\
&+ 3{\lambda_{HS_1}^{(A)}}^2\\
&+ 2{\lambda_{HS_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & \lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 8\lambda_{S_1}^{(A)}) \\
& + \mathrm{lambdaT3GeneralQ001300I1}\left(\frac{32}{3}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) + \frac{352\,(\mathrm{lambdaT3GeneralQ001300I1})^*}{3}\right) \\
& + \frac{32}{3}\,(\mathrm{lambdaT3GeneralQ001300I1})^*\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big) \\
& - \frac{248}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) \\
& - \frac{248}{45}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\ (y_1)^\dagger\Big) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) \\
& + \frac{8}{3}\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^* \\
& + \frac{832}{3}\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{lambdaT3GeneralQ000400I1})^* \\
& + \frac{16}{3}\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^* \\
& + 22{g_2}^4 \\
& - 24{g_2}^2\lambda_{S_1}^{(1)} \\
& + 2{\lambda_{12}^{(1)}}^2 \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 14{\lambda_{S_1}^{(1)}}^2 \\
& + 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & 8\lambda_{S_2}^{(1)}T_\nu^{(2)} \\
& - 8\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 24\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^* \\
& + 12{g_1}^4 \\
& - 12{g_1}^2\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 10{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_1}^{(A)}) \\
& + \mathrm{lambdaT3GeneralQ110200I1}(4\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) + 4(5\,(\mathrm{lambdaT3GeneralQ001300I1})^* + 4\,(\mathrm{lambdaT3Gene
\end{aligned}
$$

$$
\begin{aligned}
&+ 4\left(\text{lambdaT3GeneralQ110200I1}\right)^{*}\operatorname{Tr}\Big(\text{y1Conj}\ \left(y_1\right)^{\dagger}\Big)\\
&- 2\operatorname{Tr}\Big(y_1\ \left(y_1\right)^{\dagger}\ Y_e\ \left(Y_e\right)^{\dagger}\Big)\\
&- 2\operatorname{Tr}\Big(\text{y1Conj}\ \left(\text{y1Conj}\right)^{\dagger}\ Y_e\ \left(Y_e\right)^{\dagger}\Big)\\
&+ 4\lambda_5\left(\lambda_5\right)^{*}\\
&+ 20\text{lambdaT3GeneralQ001300I1}\left(\text{lambdaT3GeneralQ110200I1}\right)^{*}\\
&+ 4\text{lambdaT3GeneralQ021010I1}\left(\text{lambdaT3GeneralQ021010I1}\right)^{*}\\
&+ 6{g_2}^{4}\\
&- \frac{33{g_2}^{2}\lambda_{HS_1}^{(1)}}{2}\\
&- \frac{3{g_1}^{2}\lambda_{HS_1}^{(1)}}{2}\\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}\\
&+ 6\lambda_1\lambda_{HS_1}^{(1)}\\
&+ 4{\lambda_{HS_1}^{(1)}}^{2}\\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}\\
&+ 6{\lambda_{HS_1}^{(A)}}^{2}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2\frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = \lambda_{HS_2}^{(1)}(&4T_\nu^{(2)} + 6\operatorname{Tr}\Big(Y_d\ \left(Y_d\right)^{\dagger}\Big) + 2\operatorname{Tr}\Big(Y_e\ \left(Y_e\right)^{\dagger}\Big) + 6\operatorname{Tr}\Big(Y_u\ \left(Y_u\right)^{\dagger}\Big))\\
&- 4\operatorname{Tr}\Big(y_2\ \left(y_2\right)^{\dagger}\ Y_e\ \left(Y_e\right)^{\dagger}\Big)\\
&+ 12\lambda_5\left(\lambda_5\right)^{*}\\
&+ 12\text{lambdaT3GeneralQ110200I1}\left(\text{lambdaT3GeneralQ000211I1}\right)^{*}\\
&+ 12\text{lambdaT3GeneralQ000211I1}\left(\text{lambdaT3GeneralQ110200I1}\right)^{*}\\
&+ 12\text{lambdaT3GeneralQ021010I1}\left(\text{lambdaT3GeneralQ021010I1}\right)^{*}\\
&- \frac{9{g_2}^{2}\lambda_{HS_2}^{(1)}}{2}\\
&+ 3{g_1}^{4}\\
&- \frac{15{g_1}^{2}\lambda_{HS_2}^{(1)}}{2}\\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}\\
&+ 6\lambda_1\lambda_{HS_2}^{(1)}\\
&+ 4{\lambda_{HS_2}^{(1)}}^{2}\\
&+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2\frac{d\lambda_{12}^{(1)}}{d\ln\mu} = \lambda_{12}^{(1)}(&2T_\nu^{(1)} + 2\operatorname{Tr}\Big(\text{y1Conj}\ \left(\text{y1Conj}\right)^{\dagger}\Big) + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})\\
&+ \text{lambdaT3GeneralQ000211I1}(4\operatorname{Tr}\Big(y_1\ \left(\text{y1Conj}\right)^{\dagger}\Big) + 4(4\left(\text{lambdaT3GeneralQ000211I1}\right)^{*} + 5\left(\text{lambdaT3Gener}\right.\\
&+ 4\left(\text{lambdaT3GeneralQ000211I1}\right)^{*}\operatorname{Tr}\Big(\text{y1Conj}\ \left(y_1\right)^{\dagger}\Big)
\end{aligned}
$$

$$
\begin{aligned}
&- 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ y_2\ (y_2)^\dagger\Big)\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 20\mathrm{lambdaT3GeneralQ001300I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^*\\
&+ 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*\\
&- 12{g_2}^2\lambda_{12}^{(1)}\\
&- 6{g_1}^2\lambda_{12}^{(1)}\\
&+ 4{\lambda_{12}^{(1)}}^2\\
&+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}\\
&+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_5}{d\ln\mu} = {}&\lambda_5(T_\nu^{(1)} + \mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - 4\lambda_{HS_1}^{(A)})\\
&+ 2\,(\mathrm{lambdaT3GeneralQ021010I1})^*\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big)\\
&+ 4\,(\mathrm{lambdaT3GeneralQ021010I1})^*\,((\mathrm{lambdaT3GeneralQ000211I1})^* + 2\,(\mathrm{lambdaT3GeneralQ110200I1})^*)\\
&- \frac{21{g_2}^2\lambda_5}{2}\\
&- \frac{9{g_1}^2\lambda_5}{2}\\
&+ 2\lambda_{12}^{(1)}\lambda_5\\
&+ 2\lambda_1\lambda_5\\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5\\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = {}&\lambda_{HS_1}^{(A)}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)})\\
&- \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&+ \frac{4}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- \frac{8}{3}\lambda_5\,(\lambda_5)^*\\
&+ \frac{8}{3}\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^*\\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}\\
&- \frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}\\
&+ 2\lambda_1\lambda_{HS_1}^{(A)}\\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}\\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = \lambda_{S_1}^{(A)}(4T_\nu^{(1)} + 4\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big))$$
$$+ \mathrm{lambdaT3GeneralQ001300I1}\left(-\frac{4}{3}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\Big) - \frac{44\,(\mathrm{lambdaT3GeneralQ001300I1})^*}{3}\right)$$
$$- \frac{4}{3}\,(\mathrm{lambdaT3GeneralQ001300I1})^*\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big)$$
$$+ \frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big)$$
$$+ \frac{76}{45}\,\mathrm{Tr}\Big(y_1\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\ (y_1)^\dagger\Big)$$
$$- \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big)$$
$$- \frac{2}{3}\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\ \mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big)$$
$$- \frac{4}{3}\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^*$$
$$- \frac{224}{3}\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{lambdaT3GeneralQ000400I1})^*$$
$$- \frac{8}{3}\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^*$$
$$- 2{g_2}^4$$
$$- 24{g_2}^2\lambda_{S_1}^{(A)}$$
$$+ 3{\lambda_{HS_1}^{(A)}}^2$$
$$+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$$
$$+ 20{\lambda_{S_1}^{(A)}}^2$$

$y1Conj$

$$16\pi^2 \frac{dy1Conj}{d\ln\mu} = 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big)\,(y_1)_{ij}$$
$$+ T_\nu^{(1)}\,(\mathrm{y1Conj})_{ij}$$
$$+ \mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big)\,(\mathrm{y1Conj})_{ij}$$
$$+ \frac{1}{12}(29\left(y_1\ (y_1)^\dagger\ \mathrm{y1Conj}\right)_{ij} + 29\left(\mathrm{y1Conj}\ (y_1)^\dagger\ y_1\right)_{ij} + 6(\Big(y_2\ (y_2)^\dagger\ \mathrm{y1Conj}\Big)_{ij} + \Big(Y_e\ (Y_e)^\dagger\ \mathrm{y1Conj}\Big)_{ij} + 3$$
$$- \frac{9}{2}{g_2}^2\,(\mathrm{y1Conj})_{ij}$$
$$- \frac{3}{2}{g_1}^2\,(\mathrm{y1Conj})_{ij}$$

$lambdaT3GeneralQ000211I1$

$$16\pi^2 \frac{dlambdaT3GeneralQ000211I1}{d\ln\mu} = \lambda_{12}^{(1)}(2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\Big) + 8\mathrm{lambdaT3GeneralQ000211I1} + 10\mathrm{lambdaT3GeneralQ001}$$
$$+ \mathrm{lambdaT3GeneralQ000211I1}(2T_\nu^{(1)} + 2\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (\mathrm{y1Conj})^\dagger\Big) + 4T_\nu^{(2)})$$
$$- 4\,\mathrm{Tr}\Big(\mathrm{y1Conj}\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$$

$$
\begin{aligned}
&+ 4\,(\lambda_5)^*\,(\text{lambdaT3GeneralQ021010I1})^* \\
&+ 40\text{lambdaT3GeneralQ000400I1}\,(\text{lambdaT3GeneralQ000211I1})^* \\
&- 12{g_2}^2\text{lambdaT3GeneralQ000211I1} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ000211I1} \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ110200I1} \\
&+ 2\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000211I1} \\
&- 4\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ000211I1} \\
&+ 4\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000211I1}
\end{aligned}
$$

$lambdaT3GeneralQ000400I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ000400I1}{d\ln\mu} &= \text{lambdaT3GeneralQ000400I1}(4T_\nu^{(1)} + 4\,\text{Tr}\Big(\text{y1Conj}\ (\text{y1Conj})^\dagger\Big)) \\
&\quad + 2\text{lambdaT3GeneralQ001300I1}\,\text{Tr}\Big(\text{y1Conj}\ (y_1)^\dagger\Big) \\
&\quad - \frac{23}{15}\,\text{Tr}\Big(\text{y1Conj}\ (y_1)^\dagger\ \text{y1Conj}\ (y_1)^\dagger\Big) \\
&\quad - 24{g_2}^2\text{lambdaT3GeneralQ000400I1} \\
&\quad + 12\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000400I1} \\
&\quad - 8\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ000400I1} \\
&\quad + \text{lambdaT3GeneralQ000211I1}^2 \\
&\quad + 11\text{lambdaT3GeneralQ001300I1}^2 \\
&\quad + 2\text{lambdaT3GeneralQ110200I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ001300I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ001300I1}{d\ln\mu} &= \lambda_{S_1}^{(1)}(2\,\text{Tr}\Big(\text{y1Conj}\ (y_1)^\dagger\Big) + 22\text{lambdaT3GeneralQ001300I1}) \\
&\quad + \text{lambdaT3GeneralQ000400I1}(8\,\text{Tr}\Big(y_1\ (\text{y1Conj})^\dagger\Big) + 88\,(\text{lambdaT3GeneralQ001300I} \\
&\quad + \text{lambdaT3GeneralQ001300I1}(4T_\nu^{(1)} + 4\,\text{Tr}\Big(\text{y1Conj}\ (\text{y1Conj})^\dagger\Big)) \\
&\quad - \frac{46}{15}\,\text{Tr}\Big(y_1\ (y_1)^\dagger\ \text{y1Conj}\ (y_1)^\dagger\Big) \\
&\quad - \frac{46}{15}\,\text{Tr}\Big(\text{y1Conj}\ (y_1)^\dagger\ \text{y1Conj}\ (\text{y1Conj})^\dagger\Big) \\
&\quad - 24{g_2}^2\text{lambdaT3GeneralQ001300I1} \\
&\quad + 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ000211I1} \\
&\quad + 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ110200I1}
\end{aligned}
$$

$lambdaT3GeneralQ021010I1$

$$
\begin{aligned}
16\pi^2\,\frac{dlambdaT3GeneralQ021010I1}{d\ln\mu} &= 2\,(\lambda_5)^*\,\text{Tr}\Big(y_1\ (\text{y1Conj})^\dagger\Big) \\
&\quad + \text{lambdaT3GeneralQ021010I1}(T_\nu^{(1)} + \text{Tr}\Big(\text{y1Conj}\ (\text{y1Conj})^\dagger\Big) + 2T_\nu^{(2)} + 6\,\text{Tr}\Big(Y_d\ (Y_d) \\
&\quad + 4\,(\lambda_5)^*\,((\text{lambdaT3GeneralQ000211I1})^* + 2\,(\text{lambdaT3GeneralQ110200I1})^*) \\
&\quad - \frac{21{g_2}^2\text{lambdaT3GeneralQ021010I1}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{9{g_1}^2 \text{lambdaT3GeneralQ021010I1}}{2} \\
&+ 2\lambda_{12}^{(1)} \text{lambdaT3GeneralQ021010I1} \\
&+ 2\lambda_1 \text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_1}^{(1)} \text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_1}^{(A)} \text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_2}^{(1)} \text{lambdaT3GeneralQ021010I1}
\end{aligned}
$$

$lambdaT3GeneralQ110200I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ110200I1}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)} (2 \operatorname{Tr}\left(\text{y1Conj } (y_1)^\dagger\right) + 10\text{lambdaT3GeneralQ001300I1} + 8\text{lambdaT3GeneralQ11} \\
& + \text{lambdaT3GeneralQ110200I1} (2T_\nu^{(1)} + 2 \operatorname{Tr}\left(\text{y1Conj } (\text{y1Conj})^\dagger\right) + 6 \operatorname{Tr}\left(Y_d \ (Y_d)^\dagger\right) + \\
& - 2 \operatorname{Tr}\left(\text{y1Conj } (y_1)^\dagger \ Y_e \ (Y_e)^\dagger\right) \\
& + 4 (\lambda_5)^* (\text{lambdaT3GeneralQ021010I1})^* \\
& + 40\text{lambdaT3GeneralQ000400I1} (\text{lambdaT3GeneralQ110200I1})^* \\
& - \frac{33{g_2}^2 \text{lambdaT3GeneralQ110200I1}}{2} \\
& - \frac{3{g_1}^2 \text{lambdaT3GeneralQ110200I1}}{2} \\
& + 6\lambda_1 \text{lambdaT3GeneralQ110200I1} \\
& + 2\lambda_{HS_2}^{(1)} \text{lambdaT3GeneralQ000211I1} \\
& + 2\lambda_{S_1}^{(1)} \text{lambdaT3GeneralQ110200I1} \\
& - 4\lambda_{S_1}^{(A)} \text{lambdaT3GeneralQ110200I1}
\end{aligned}
$$

$muT3GeneralS1Sq$

$$
\begin{aligned}
16\pi^2 \frac{dmuT3GeneralS1Sq}{d\ln\mu} = {} & m_1^2 (2 \operatorname{Tr}\left(\text{y1Conj } (y_1)^\dagger\right) + 10\text{lambdaT3GeneralQ001300I1}) \\
& - 4 \operatorname{Tr}\left(M_F \ (M_F)^\dagger \ (\text{y1Conj})^T \ (y_1)^*\right) \\
& + \text{muT3GeneralS1Sq} (2T_\nu^{(1)} + 2 \operatorname{Tr}\left(\text{y1Conj } (\text{y1Conj})^\dagger\right)) \\
& + 40\text{lambdaT3GeneralQ000400I1} (\text{muT3GeneralS1Sq})^* \\
& - 12{g_2}^2 \text{muT3GeneralS1Sq} \\
& + 2\lambda_{S_1}^{(1)} \text{muT3GeneralS1Sq} \\
& - 4\lambda_{S_1}^{(A)} \text{muT3GeneralS1Sq} \\
& + 2\text{lambdaT3GeneralQ000211I1} m_2^2
\end{aligned}
$$